

Research Article

Experimental and DFT TB-mBJ calculations studies of structural, morphological, electronic, optical and electrical properties of copper oxide thin films

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ABSTRACT

The present work reviews a study of copper oxide (CuO) thin film properties elaborated via two distinct techniques: spin-coating and sprays pyrolysis. The primary objective of the work is to investigate the experimental results and theoretical finding by DFT TB-mBJ calculations of pure CuO films. First of all, the structural properties were studied by X-ray diffraction; Where the CuO films show polycrystalline and monoclinic structures without any other phases. Furthermore, Field emission scanning electron microscopy (FESEM) investigates the effects of elaboration methods on the morphology and surface composition and also, the film thickness was about 234,1 nm for sprayed film and 449 nm for spin-coated films. Moreover, the Optical analysis exhibits a higher absorbance for sprayed films than the spin-coated ones where the band gap energy was found 1.25 eV for using spin-coated and 1.35 eV for sprayed thin films respectively. Furthermore, the electrical conductivity of the spin coated films is higher than the sprayed ones. The band structure, density of states (DOS), electronic density and optical properties of pure CuO were estimated using the LDA approximation in conjunction with the TB-mBJ approximation in order to investigate its characteristics. These results indicate that the CuO films are suitable to be used as an absorber of thin-film solar cells.

1. Introduction

Nowadays, solar energy is considered to be a very important source of energy to meet the future requirement of energy without penalizing the environment [1–3]. The scientific community gives great attention to develop suitable materials with high efficiency for photovoltaic applications [2]. Metal oxide is exploited as the most conventional material for solar cells [4,5]. Copper oxide (CuO) is among the most fascinating transition metal oxide semiconductors marked with monoclinic structure and a p-type conductivity [3–8]. The band gap values of CuO films are in the range of 1.2–2.1 eV with high charge carrier mobility, which can absorb most of visible light [8,9]. Moreover, CuO has attracted the attention due to its economic availability, non-toxicity and

optoelectronic properties [9]. Also, CuO is one of the best choices for solar energy applications. In theory, its conversion efficiency can reach 33% for a single junction PV cell with a CuO band gap of 1.4 eV [10]. However, in the experimental results, the conversion efficiency of CuO/Si based on heterojunction PV cells does not exceed 0.41% [11]. CuO is a selective choice in many applications including gas sensing, magnetic storage media, photocatalysis, photoelectrochemical cells, and batteries [12–14].

Thin films of CuO were synthesized by several methods, for example spray pyrolysis [14], spin-coating [15], Chemical Solution Deposition (CSD) method [16], Magnetron Sputtering technique [17], Pulse Laser Deposition [18] and Sequential Ionic Layer Adsorption Reaction [19].

Theoretical investigations are crucial in addition to experimental

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studies since they will yield prior knowledge of the properties of the materials. One of the tried-and-true methods for precisely adjusting material properties and approving experimental findings is a DFT study. This is how the LDA and TB-mbj approximations are applied in this work, and we can draw the conclusion that the TB-mbj approximations are in good agreement with the experimental research to support the characteristics of CuO semiconductors and confirm in theory their excellent photovoltaic performance. In this work we calculate the band structure, the density of states and optical properties of CuO.

In order to integrate into the CuO lattice structure and carefully examine its structural, optical, and electrical properties, we have decided to use spray pyrolysis and the sol-gel spin coating approach in this study. In this regard, we looked at how the elaboration procedure affected the CuO thin film. Both methods were chosen due to their superior qualities. Spin-coating is a simple technique for forming with low cost and low processing temperature the uniform films with precise stoichiometry over large areas [20] and Spray pyrolysis is advantageous due to its low cost, chemical composition control and controllable thickness [21].

In addition to a theoretical investigation using LDA and TB-mbj approximations, there hasn't been any research on CuO thin films using two different techniques of elaboration to investigate the structural and optical properties using experimental methods (XRD-Diffraction, UV-visible characterization, FE-SEM with EDX analysis, and electrical properties).

2. Materials and methods

2.1. Procedure for spray pyrolysis

A low-cost spray pyrolysis method was used to deposit CuO thin films. As a precursor, 0.2 M of dehydrated copper chloride ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$) was dissolved in 45 ml of deionized water. The solution was stirred for 15 min at room temperature. The ethanol and distilled water were used just before the deposition to eliminate any dirt particles or contaminants from the surface. The solution was manually sprayed on the pre-heated glass substrate at a temperature of 300 °C using a perfume nozzle that was fixed perpendicular at a distance of 20 cm. The sprayed time was varied between 6 and 8 min to investigate the effect of time on the properties of CuO. Subsequently, sprayed films underwent thermal treatment for 1 h at 300 °C.

2.2. Procedure for spin-coating

Copper oxide thin film was synthesized by sol-gel spin-coating method. 0.2 M of dehydrated copper chloride ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$) was dissolved in ethanol, with drops of Monoethanolamine (MEA) added as a stabilizer with a $[\text{MEA}]/[\text{Cu}]$ molar ratio of 1. The glass substrate was cleaned with ethanol and distilled water. After 15 min of continuous stirring at room temperature, a clear blue sol-gel was obtained. Spin-coating speed and time were 3000 r/min and 40 s respectively. Every layer was dried for 5 min on the hotplate at 150 °C. This coating process was repeated between eight and sixteen times to obtain pure CuO films with varying thickness layers. Finally, the CuO films were annealed in air atmosphere furnace at 450 °C for 1 h.

2.3. Characterization

X-ray diffractometer (XRD) by diffractometer of the "BRUKER eco" type (Θ - Θ) configuration and using CuK_α radiation (1.541 Å) were used to characterize the prepared thin films. Field emission scanning electron microscopy was used to study the morphology and microstructural compositions (FESEM). A JEOL EX-37001 electron dispersive spectrometer attached to the FESEM was used for elemental analysis. A UV-Vis spectrophotometer was used to measure transmission and absorption over the wavelength range of 200–1100 nm. Measurements of

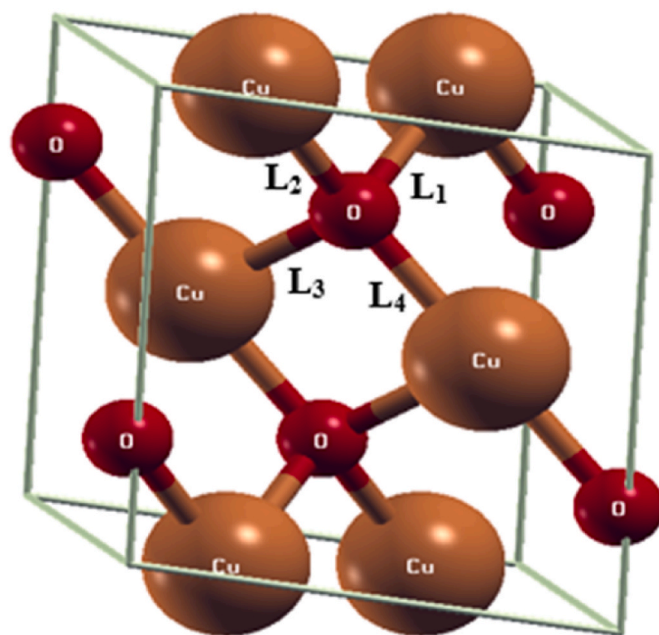


Fig. 1. The optimized structure of monoclinic CuO.

electrical resistivity were made using a JANDEL model RM3 digital multimeter at room temperature.

2.4. DFT TB-mBJ calculations

Our calculations are performed based on density functional theory as implemented in the Wien2K code [22]. We adopted the Local-density approximation (LDA) to describe the exchange and correlation functional. Since LDA failed in revealing the semiconducting nature of CuO, we introduced the Tran-Blaha modified Becke-Johnson potential (TB-mBJ potential) [23]. Due to electron self-correlation inaccuracy and an inherent absence of derivative discontinuities, the standard DFT typically provides lower band-gap values for structures. The precision of the computations is increased by determining the self-energy of an electron many-body system using the TB-mBJ approximation.

In our calculations we used the cut-off parameters of $l_{\text{max}} = 10$ and $\text{RMT}^*K_{\text{max}} = 7$ (where RMT is the smallest muffin-tin radius in the unit cell and K_{max} is the magnitude of the largest K wave vector). The muffin-

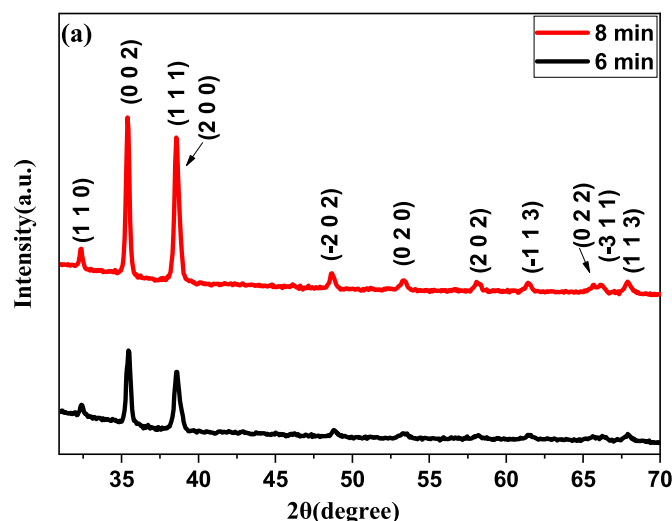


Fig. 2.a. XRD patterns of CuO thin films deposited by spray pyrolysis.

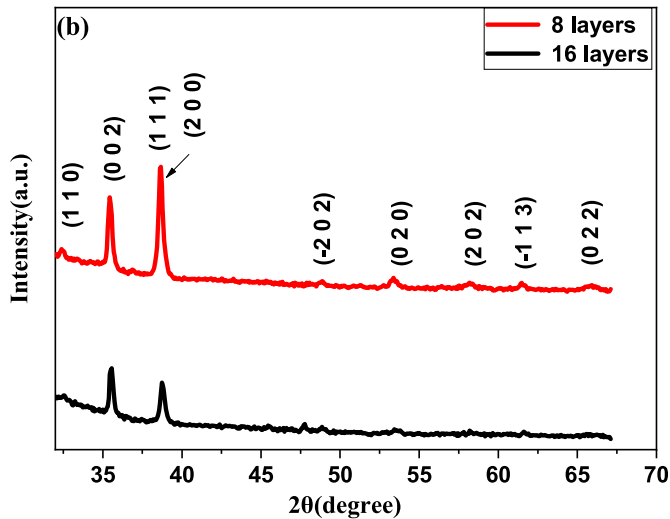


Fig. 2.b. XRD patterns of CuO thin films deposited by spin-coating.

tin sphere radii adopted for respectively, O and Cu are 1.63 and 1.72 a. u.. The results are obtained with an energy convergence criterion of 10^{-6} Ry and sampling of $[12 \times 12 \times 12]$ grid for the brillouin zone. The optimized studied structure of CuO is shown in Fig. 1. The optimized parameters are $a = 8.793$ bohr, $b = 6.444$ bohr, $c = 9.653$ bohr, $\alpha = \gamma = 90^\circ$ and $\beta = 99.48^\circ$. Using the program mini. Positions offered by the Wien2k package, which is interested in internal parameter minimization, we determined the equilibrium positions for all atoms in our structure. The obtained total energy after minimization is -13841.911 Ry. The computed Cu–O bond lengths is $L_1 = 1.918$ Å, $L_2 = 1.963$ Å, $L_3 = 1.918$ Å, and $L_4 = 1.963$ Å (Fig. 1).

3. Results and discussion

3.1. Structural studies

Fig. 2 shows the XR-diffraction patterns of CuO films elaborated by spin-coating and spray pyrolysis. Fig. 2 (a) depicts the XRD patterns of sprayed CuO thin films deposited on a glass substrate at various growth times. It is observed that the peaks become more intense with increasing deposition time, which can be justified by the growth of the film thickness [24] and also the degree of crystallinity can be correlated with the XRD peak intensities. In both samples, no additional typical copper oxide phase, like Cu_2O , was found in either of the samples. Diffraction peaks are visible in the XRD pattern at $2\theta = 32.49^\circ, 35.5^\circ, 38.78^\circ, 38.93^\circ, 48.66^\circ, 53.43^\circ, 58.37^\circ, 61.49^\circ, 65.77^\circ, 66.18^\circ$ and 67.96° corresponding respectively to (110), (002), (111), (200), (-202) , (020), (202), (-113) , (022), (-311) and (113) plans. Two main peaks are distinguished at $2\theta = 35^\circ$ and $2\theta = 38^\circ$ corresponding to the (002) and (111) reticular plans respectively.

Concerning sol gel spin-coated CuO thin films, which were deposited with 8 and 16 layers. As shown in Fig. 2 (b), the elaboration method has

an effect on the crystal phase of CuO, as shown by the absence of other peaks. The highest diffraction peaks are observed at $2\theta = 35^\circ$ and $2\theta = 38^\circ$ corresponding to the (002) and (111) plans. In addition, decreasing the number of layers increased the intensity of the peaks, which can be described by an improvement in the crystalline quality of the CuO thin films.

The XRD examination allows us to verify that the peaks observed are confirmed with (ICDD card no 00-005-0254). As it can be seen, the Copper oxide films are polycrystalline and crystallize in monoclinic structure [25,26].

Using the most intense peaks of the X-Ray diffraction, the microstructural parameters (Table 1) such as crystallite size (D), dislocation density (δ), crystalline number (N), and microstrain (ϵ) were calculated [12].

$$D = \frac{k\lambda}{\beta \cos \theta} \quad (1)$$

$$\delta = \frac{1}{D^2} \quad (2)$$

$$N = \frac{d}{D^3} \quad (3)$$

$$\epsilon = \frac{\beta \cos \theta}{4} \quad (4)$$

With.

λ : X-ray wavelength,

β : The full-width at half-maximum (FWHM) of the diffraction peaks,

θ : The diffraction angle.

To investigate the microstructural characteristics, the crystallite size increases after bumping up the deposition time and number of layers, as shown in Table 1, in the other hand the dislocation density, microstrain, and crystallite number decrease. Usually, the small crystallites presented on the samples elaborated by the spin coating method. The increase in D-value could be attributed to a reduction in grain boundaries and, as a result, the number of defects in the films. This difference in structural characteristics could be caused by an improvement of the crystallinity [27].

3.2. FE-SEM with EDX analysis

Following Fig. 3 illustrates the surface morphology of CuO thin films prepared via spin-coating and spray pyrolysis methods. Fig. 3(a) shows that the sprayed film has a dense and uniform shape where the nanoparticles are smooth and uniformly distributed on the glass substrate. The structure of CuO film has been confirmed by FESEM micrographs that are consistent with the literature [8,27–29]. Fig. 3(b) depicts the surface morphology of the spin-coated CuO thin film, which reveals a difference in growth with improved texture, a compact structure, and more distinct granule boundaries. This type of creation has been reported in the literature as well [8,29]. The average crystallite size on the surface increases from 73.98 nm to 146 nm when the deposition method is changed from spray pyrolysis to spin-coating, this Changes may be

Table 1
Microstructural parameters of CuO thin films.

	sample	hkl	FWHM	D (nm)	$\delta (\times 10^{-3} \text{ nm}^{-2})$	$\epsilon (\times 10^{-3})$	$N (\times 10^{16} / \text{m}^2)$
Spray pyrolysis	6 min	0 0 2	0.31	25.71	1.51	1.4	1.37
		1 1 1	0.36	22.07	2.05	1.63	2.17
	8 min	0 0 2	0.298	29.21	1.17	1.23	0.93
		1 1 1	0.34	25.85	1.49	1.39	1.35
Spin-coating	8 layers	0 0 2	0.37	23.55	1.8	1.5	3.44
		1 1 1	0.34	25.65	1.51	1.41	2.66
	16 layers	0 0 2	0.337	25.85	1.49	1.39	2.6
		1 1 1	0.306	28.73	1.2	1.25	1.89

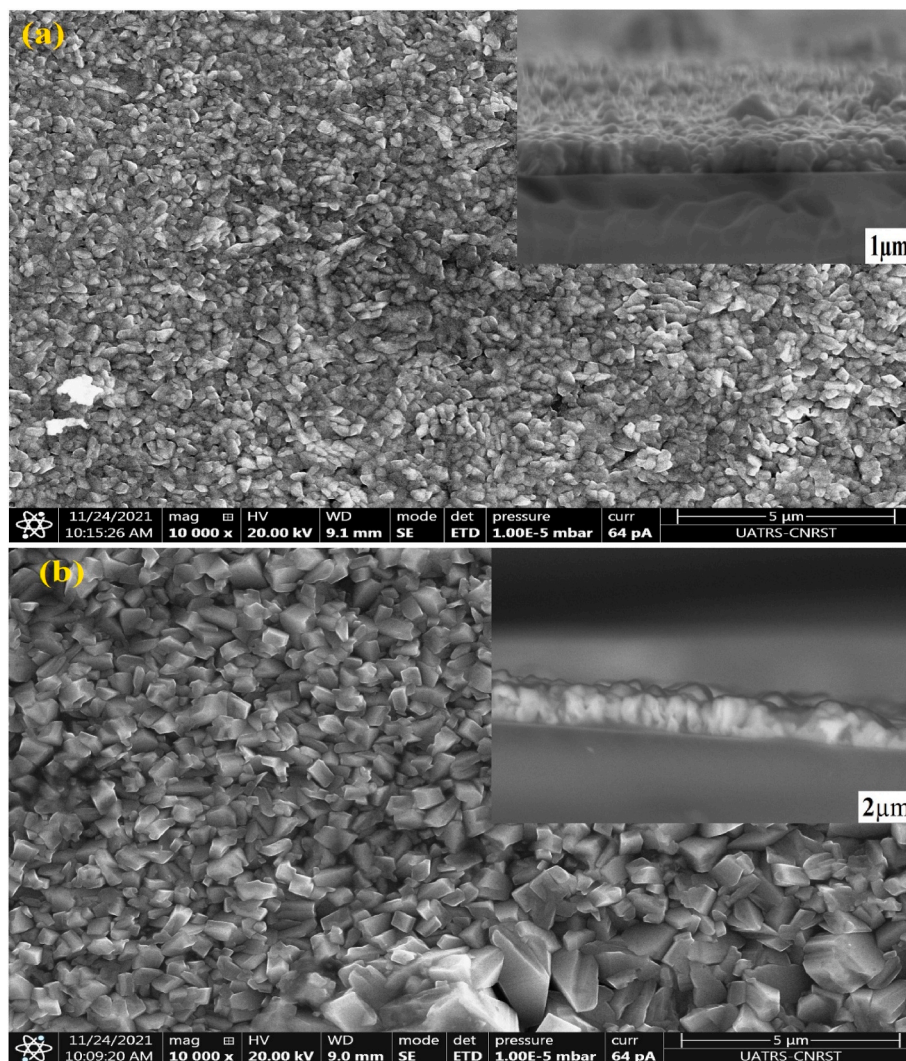


Fig. 3. Morphology images of CuO films elaborated by (a) spray pyrolysis (6 min) and (b) spin-coating (8 layers).

Table 2
Thin films thickness of CuO.

			Thickness (nm)
Spray pyrolysis	6 min		234.1
	8 min		292
Spin-coating	8 layers		449
	16 layers		898

connected to strain that develops in the crystal lattice [30]. The films thickness was estimated according to the cross-section images and shown in Table 2.

EDX analysis was performed on various areas of the substrate in order to validate the creation of CuO thin films. Fig. 4 depicts the EDX analysis spectrum of CuO thin films. The EDX spectrum revealed two strong peaks corresponding to Cu and O, confirming that the resulting films were close to stoichiometry. An additional peak corresponding to Si was noted very near the Cu peak, which was attributed to the glass substrate's composition.

The atomic and weight percent % for different chemical elements in CuO thin films are identified. The EDS spectrum of the pulverized CuO sample revealed that the weight percentages (wt%) of Cu and O are 66.26 and 22.14, respectively. In the spectrum of the spinning film, the weight percentage of Cu and O are respectively 75.3 and 19.47. The

outcomes are very related to literature [31,32].

3.3. Optical properties

Fig. 5 shows the absorbance spectra of thin films CuO in the wavelength range of 300–900 nm, where the deposition was by spin-coating and spray pyrolysis methods. The optical absorbance is high in the visible range while because of the arriving photon energy is above or equal to the band gap of the semiconductor [24,26]. As illustrated in Fig. 5 (a) the absorbance spectra of the sprayed films are similar to those in Ref. [12] where the deposition time affects the absorbance of the films. These results were found to be due to less dispersion as the degree of irregularity in the grain size distribution decreases [29]. As illustrated in Fig. 5 (b) the absorption increases by decreasing the number of layers, this increase may be related to an advancement in films thickness. Overall, this can be explained by the increased photon scattering caused by crystal defects due to a difference in layer number [33]. In addition, drops in absorption values are observed near infrared region.

We can conclude that, when the value of the incoming photon energy is greater than or equal to the semiconductor gap, CuO thin films show a high absorption values in the visible range. This gives the CuO thin film an absorption property that is advantageous for photodetection of visible light [24,34].

The band gap energy (E_g) of copper oxide thin films was determined by Tauc's relation [34].

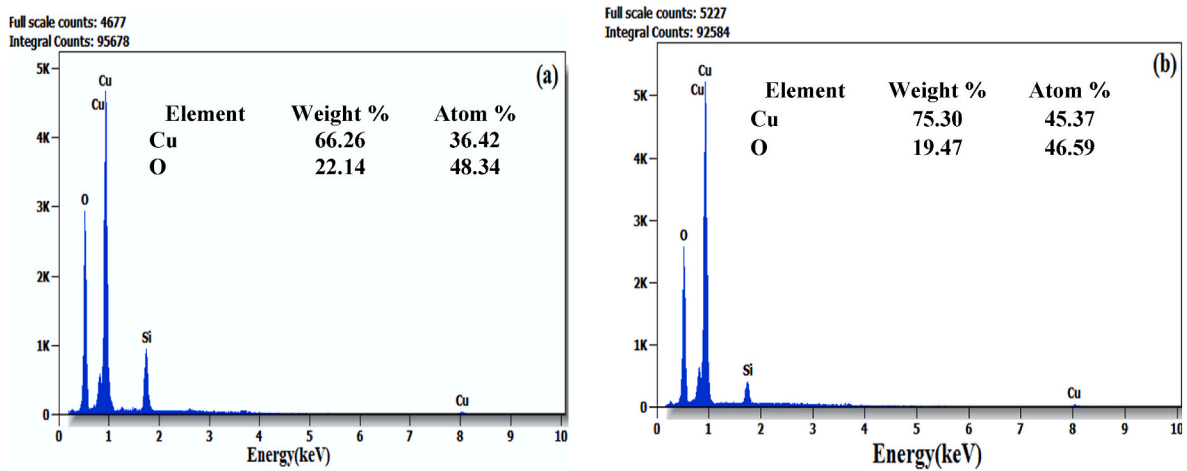


Fig. 4. EDX spectrum of (a) sprayed (6 min) (b) spin coated (8 layers) CuO thin films.

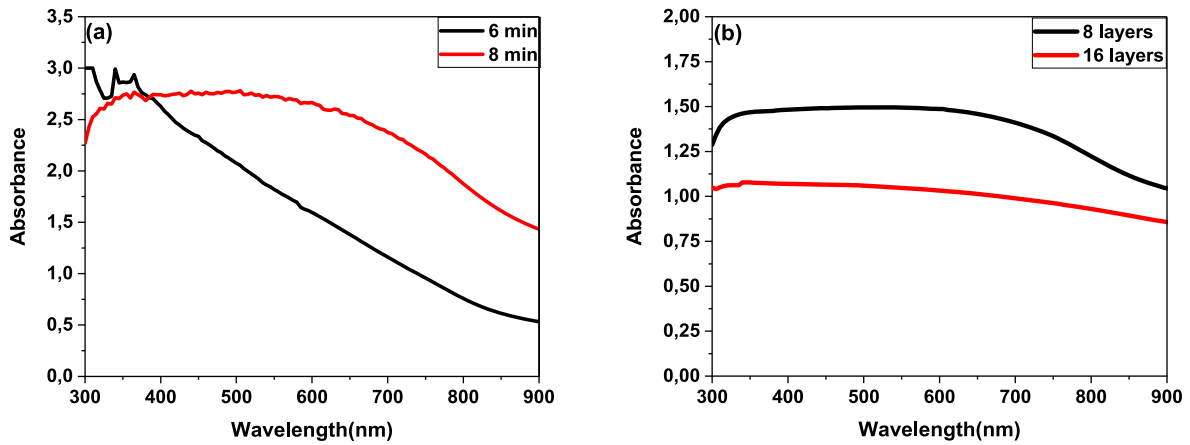


Fig. 5. Absorbance for CuO thin films elaborated by (a) spray pyrolysis and (b) spin-coating.

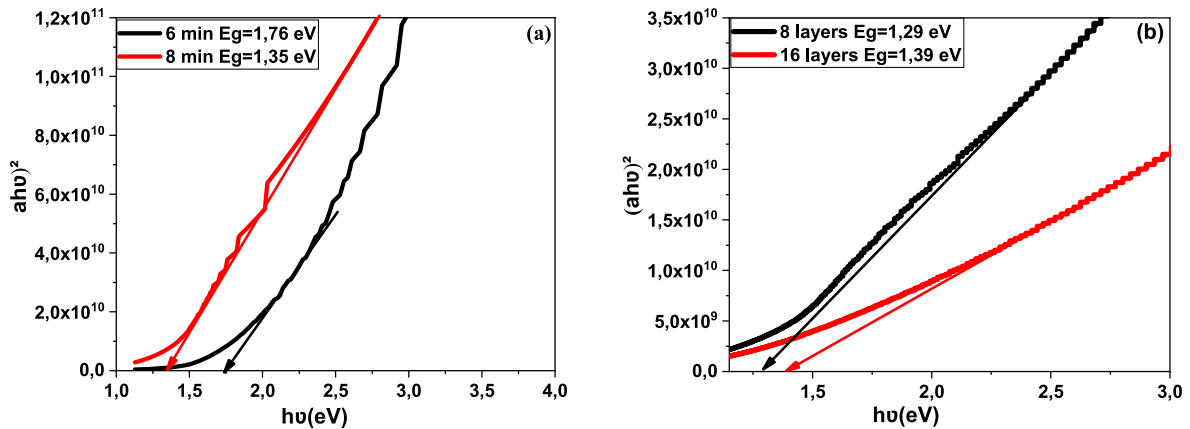


Fig. 6. CuO's optical band gap energy deposited by (a) spray pyrolysis and (b) spin-coating.

$$(\alpha h\nu)^2 = A (h\nu - E_g) \quad (5)$$

E_g is the optical band gap and A is a characteristic parameter independent of photon energy. Fig. 6 shows $(\alpha h\nu)^2 = f(h\nu)$ for the determination of the direct band gap of the CuO semiconductor by extrapolating the linear part of the graph. Besides that, the alteration in E_g could be caused by the CuO nanostructures' changed grain sizes [35]. As the number of layers in spin-coated thin films increases, so does their

band gap energy values from 1.29 eV to 1.39 eV. The increase in thickness and modifications to the morphological properties are thought to be the reasons why band gaps are shrinking in spin-coated CuO thin films [36]. For the sprayed CuO thin films, the band gap energy values are found to be 1.35 eV for a deposition time of 8 min and 1.76 eV for a growth time of 6 min. The values of the band gap are comparable to those from published research [20,37].

Our findings demonstrate extremely low transmittance, high

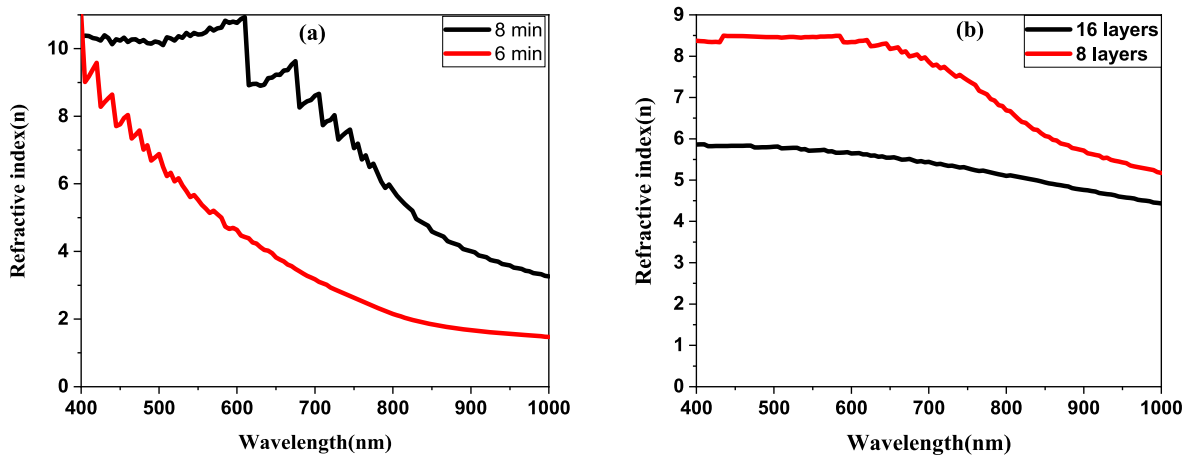


Fig. 7. Refractive index of CuO thin films deposited by (a) spray pyrolysis and (b) spin-coating.

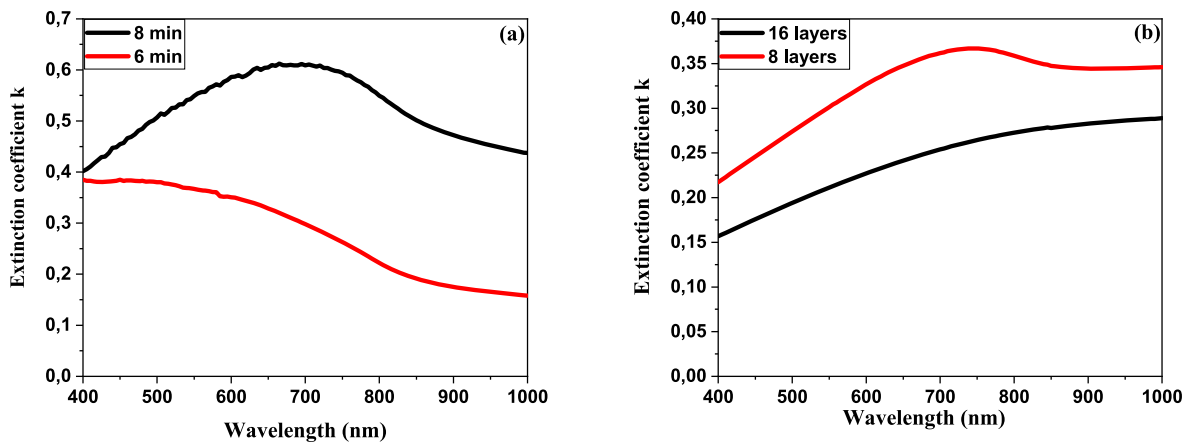


Fig. 8. Extinction coefficient of CuO thin films deposited by (a) spray pyrolysis and (b) spin-coating.

absorbance, and low band gap energy near 1.4 eV, which is in line with our hypothesis that CuO thin films would be an excellent material for the preparation of solar cells [10,38].

Refractive index and extinction coefficient are important optical constants to understanding the behavior of materials in electric fields. In an absorbing medium, the refractive index and extinction coefficient are written using the shaping equation (6) [39]:

$$n = \sqrt{\frac{4R}{(1-R)^2} - k^2 + \frac{1+R}{1-R}} \quad (6)$$

$$k = \frac{\alpha\lambda}{4\pi} \quad (7)$$

R, and α are the reflectance and the absorption coefficient, respectively, and λ is the wavelength.

Figs. 7 and 8 depict the variations in extinction coefficient and

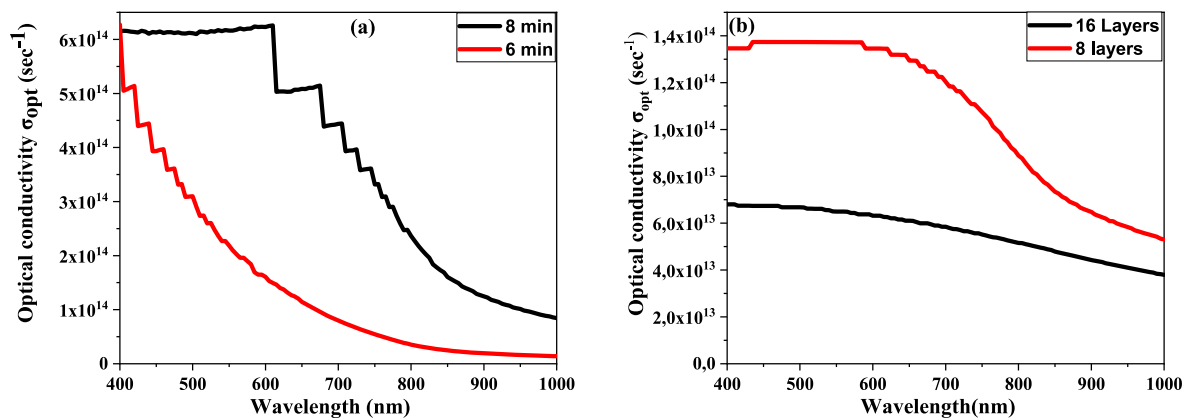


Fig. 9. Optical conductivity of CuO thin films deposited by (a) spray pyrolysis and (b) spin-coating.

Table 3

Electrical properties of CuO thin films produced by spray pyrolysis and spin coating.

	Thickness (nm)	Electrical sheet Resistance ($M\Omega/\square$)	Resistivity (Ωcm)	Conductivity(Ωcm) ⁻¹
Spray pyrolysis	234.1	12.26	2.87	0.348
Spin-coating	449	4.44	1.99	0.5

Table 4

Comparison of experimental and theoretical band gap values of pure CuO films.

Experimental band gap, Eg (eV)				Theoretical band gap, Eg (eV)
Spray pyrolysis		Spin-coating		LDA-mBJ approximation
6 min	8 min	8 layers	16 layers	1.66
1.76	1.35	1.29	1.39	

refractive index as a function of wavelength.

It has been found that as the wavelength increases, the values of n and k rise. They exhibit typical final dispersion as a result.

Fig. 7. (a) shows that the refractive index value for thin films that are sprayed is higher by increasing sprayed time in the visible range. Because the film porosity decreases with longer deposition times, this is expected. For coated thin films as illustrated in Fig. 7. (b) the highest refractive index values are seen in the wavelength range 400–800 nm, this difference may be related to an increase in packing density.

Fig. 8 illustrates the relationship between the wavelength and the extinction coefficient (k) values of films with various numbers of coated layers and deposition times. Fig. 8 (a) shows that there is a distinct absorption band with a center wavelength of 500–800 nm that grows dramatically as the sprayed time increases, which can be related to film thickness [20]. The extinction coefficient rises in Fig. 8 (b) as the coated layers decrease in relation to the absorbance. In addition, the high absorbance was blamed for the rising extinction coefficient values.

An essential material property for applications in optoelectronic devices is the optical conductivity. This optical phenomenon happens as a result of a material's reaction to an electromagnetic wave. The relationship between this parameter and the absorption coefficient is as follows [39]:

$$\sigma_{opt} = \frac{anc}{4\pi} \quad (8)$$

Fig. 9 shows the optical conductivity of CuO thin films deposited with spray pyrolysis and spin-coating techniques as a function of wavelength, the visible and near-IR spectral ranges show that all

samples have reasonably high absorption coefficients. The optical conductivity of sprayed films decreases with increasing deposition time from 6 to 8 min (Fig. 9 (a)). As presented in Fig. 9 (b) the optical conductivity of spin-coated films decreases with increasing cycles number. The obtained results are similar to the earlier finding [2]. Due to the strong optical conductivity of CuO thin films, they make a viable choice for optoelectronic applications.

3.4. Electrical properties

The four-point technique was used to measure the electrical characteristics of CuO thin films created using two different methods. The voltage will be reported by the inner probes after a current has passed through the two outer probes. Calculating the electrical sheet resistance R_s of the films is done using this formula [40]:

$$R_s = \frac{\pi}{\ln(2)} \left(\frac{V}{I} \right) \quad (9)$$

Where R_s sheet resistances are measured by the device.

The following equation has been used to calculate the electrical resistivity [41]:

$$\rho = d * R_s \quad (10)$$

Where d is the thickness of the film and the electrical conductivity has been calculated using the following equation:

$$\sigma = \frac{1}{\rho} \quad (11)$$

The results were summarized in Table 3. It is observed that the method of elaboration has an impact on the electrical properties of CuO thin films. As demonstrated, coated film has lower sheet resistance than sprayed film, which can be attributed to the effect of oxygen flow rates [33]. In addition, the outcomes showed that the electrical conductivity increased in thicker films, indicating that the spin coating technique is a promising method for achieving the highest electrical conductivity [26].

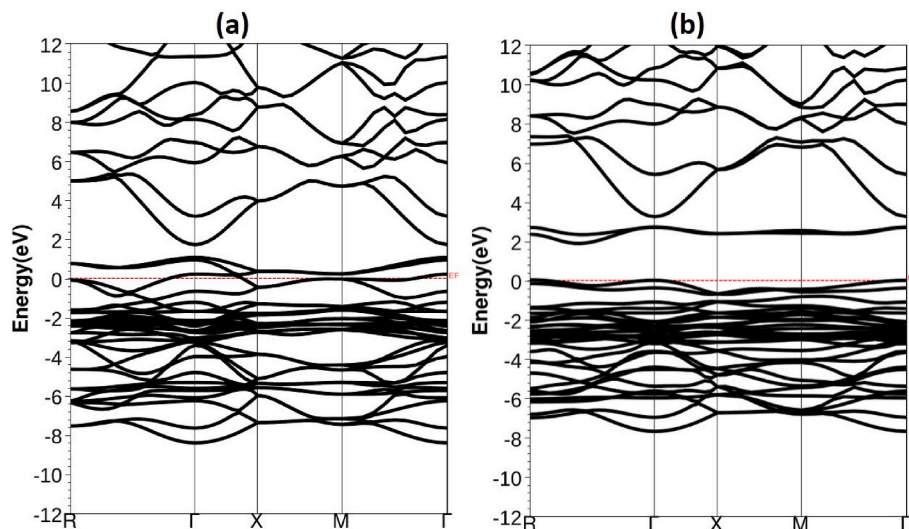


Fig. 10. Band structure of pure cupric oxide (CuO) in (a) LDA approximation and (b) LDA-mBJ approximation. The horizontal dotted line at zero indicates the Fermi energy level.

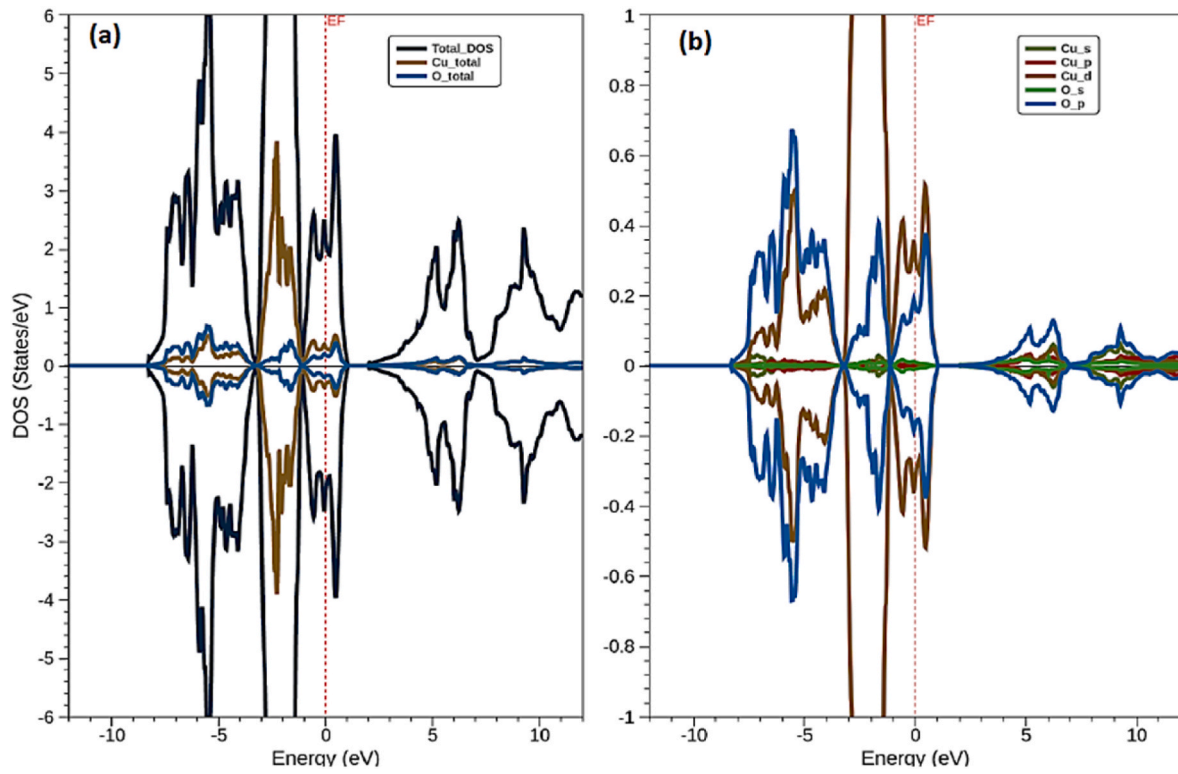


Fig. 11. Total density of states (TDOS) and partial density of states (PDOS) in CuO compound computed by LDA approximation. The vertical dotted line at zero indicates the Fermi energy level.

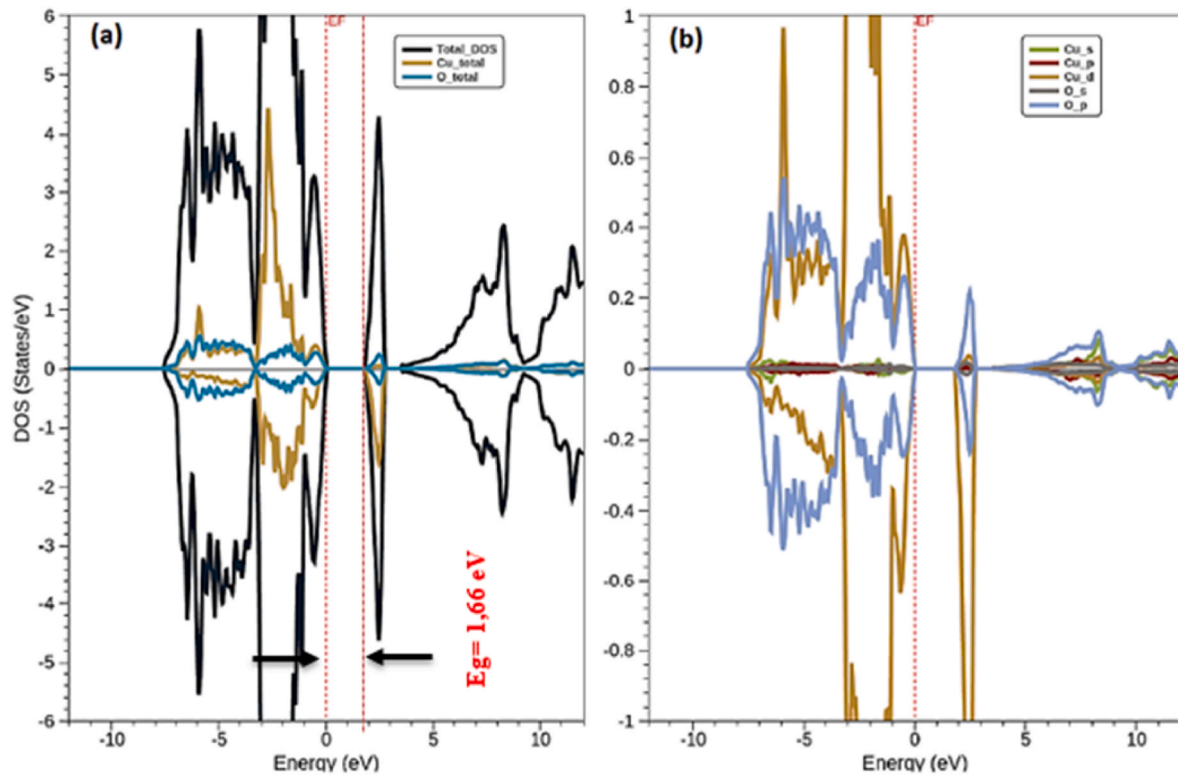


Fig. 12. (a) Total (TDOS) and (b) partial (PDOS) density of states in CuO compound computed by LDA-mBJ approximation. The vertical dotted line at zero indicates the Fermi energy level.

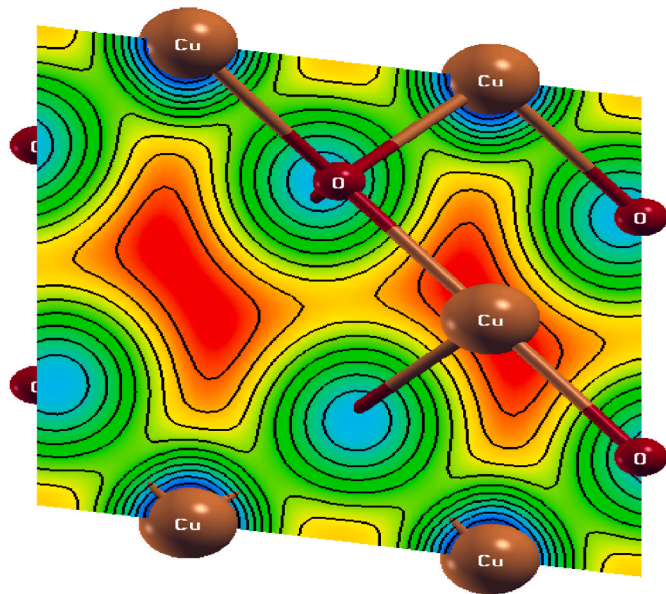


Fig. 13. Electronic density distributions of pure monoclinic CuO.

3.5. Electronic properties of CuO based on DFT TB-mBJ calculations

We estimated the band structure, density of states (DOS), and electron density distribution of pure CuO in the context of the LDA approximation combined with the TB-mBJ approximation in order to investigate its electrical characteristics.

Fig. 10(a) of the LDA calculation depicts pure CuO as having a metallic quality, which is inconsistent to earlier experimental findings [42,43]. This highlights the limitations of the LDA approach for the analysis of the CuO compound and demonstrates why the mBJ

approximation was used in this investigation. The LDA-mBJ calculated band structure, as shown in Fig. 10(b), has a direct bandgap energy of approximately 1.66 eV and a general band appearance that is extremely close to our experimental results and in line with earlier findings [44,45] (Table 4).

Figs. 11 and 12 show in the LDA and LDA-mBJ approximations, respectively, the total (TDOS) and partial (PDOS) density of states in the CuO compound.

It is obvious from Fig. 11 (a). and Fig. 12 (a). that Cu and O contribute equally to the conduction band constitution in both approximations. from Fig. 11 (b) and Fig. 12(b). The Cu element appears to contribute by d levels, whereas the O element appears to contribute by p levels. The same thing can be observed concerning the valence band. Additionally, it is possible to see the hybridization of O-2p and Cu-3d states. The O-2p and Cu-3d orbitals are said to be responsible for the creation of the CuO band gap [46]. This agrees with the electronic density distribution of Fig. 13, where we see that the Cu-O bond presents a mixed covalent and ionic character. The covalent bond results from the previously described hybridization of Cu-3d and O-2p states in the VBM, while the ionic bond results from O's high electronegativity.

3.6. Optical properties using DFT TB-mBJ calculations

The optical properties are obtained through solving the following complex dielectric function:

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \quad (12)$$

In which $\varepsilon_1(\omega)$ expresses the real part of dielectric function calculated following the Kramers-kronig transformation:

$$\varepsilon_1(\omega) = 1 + \frac{2}{\pi} P \int_0^{\infty} \frac{\omega' \varepsilon_2(\omega')}{\omega'^2 - \omega^2} d\omega' \quad (13)$$

And $\varepsilon_2(\omega)$ designates the imaginary part calculated using summation

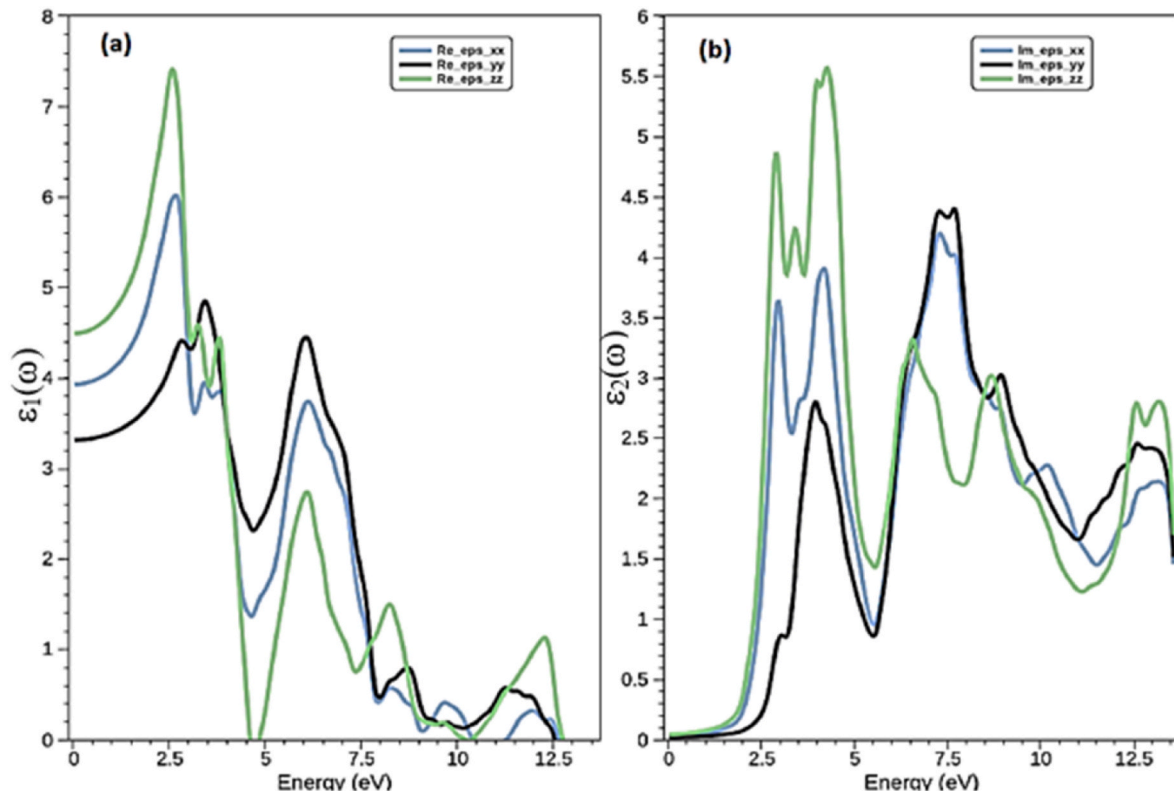


Fig. 14. (a) Real and (b) Imaginary parts of dielectric function of monoclinic CuO.

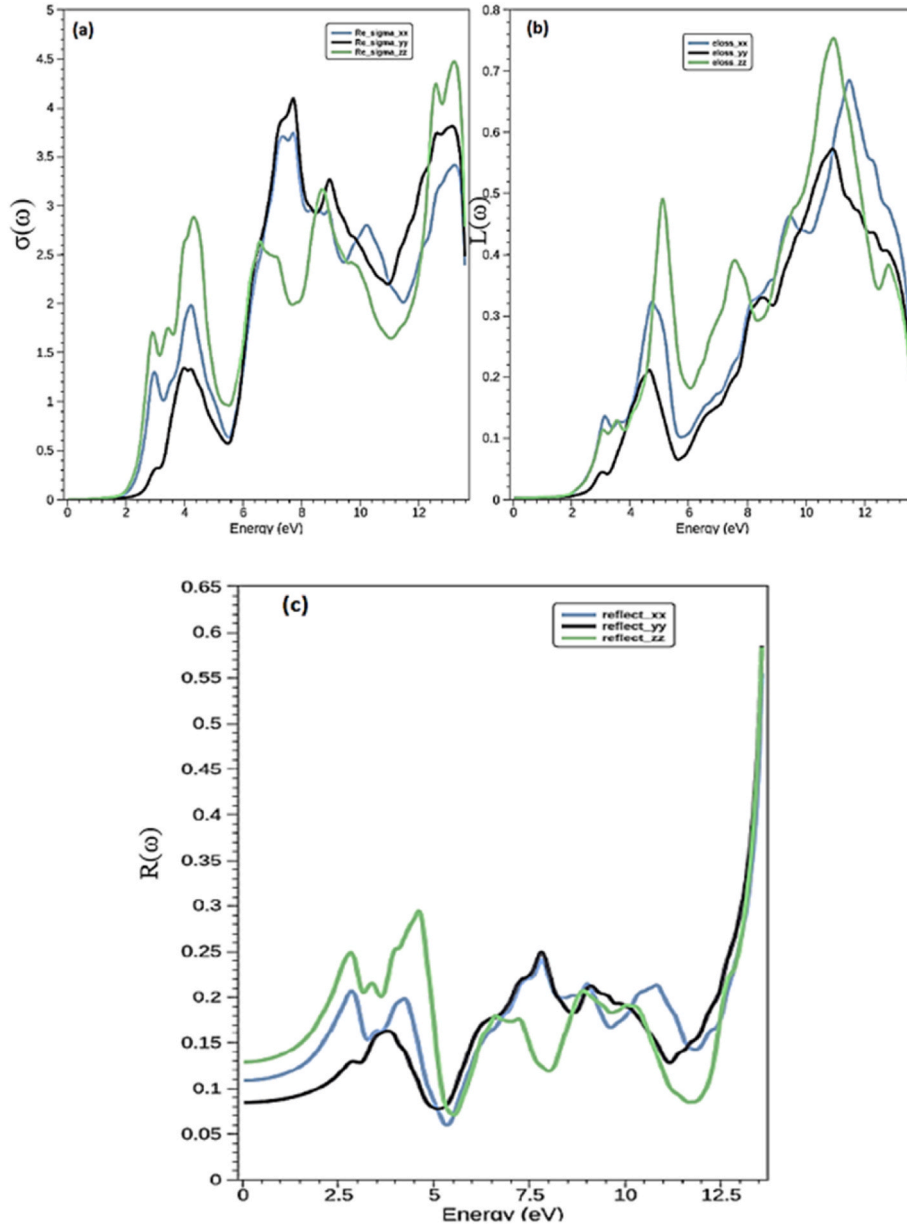


Fig. 15. The (a) optical conductivity (b) Electron energy loss spectrum and (c) reflectivity of CuO.

over the empty states:

$$\varepsilon_2(\omega) = \left(\frac{4\pi e^2}{\omega^2 m^2} \right) \sum_{ij} \int \langle i | M | j \rangle^2 f_i (1 - f_j) * \delta(E_j - E_i - \omega) d^3k \quad (14)$$

p refers to the principal value of the integral, e indicates the elementary charge, m is standing to the electron's mass, M expresses the dipole matrix element, i and j indicate to respectively, the initial and final states, f_i corresponds to the Fermi-Dirac Distribution Function of i -th state and E_i represents the energy of an electron in the i -th state with a wave number k .

Knowing $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$, other optical properties such as the optical conductivity $\sigma(\omega)$, the energy loss function the reflectivity $R(\omega)$, the refractive index and extinction coefficient $k(\omega)$ can be obtained by the formula [47,48]:

$$\sigma(\omega) = -i \frac{\omega}{4\pi} [\varepsilon(\omega) - 1] \quad (15)$$

$$L(\omega) = \varepsilon_2 / [\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)] \quad (16)$$

$$R(\omega) = \left[\frac{\sqrt{\varepsilon_1(\omega) + i\varepsilon_2(\omega)} - 1}{\sqrt{\varepsilon_1(\omega) + i\varepsilon_2(\omega)} + 1} \right]^2 \quad (17)$$

$$n(\omega) = \left[\sqrt{\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)} + \varepsilon_1(\omega) \right]^{1/2} / \sqrt{2} \quad (18)$$

$$k(\omega) = \left[\sqrt{\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)} - \varepsilon_1(\omega) \right]^{1/2} / \sqrt{2} \quad (19)$$

To reveal the optical anisotropy, the optical properties are calculated for incident radiations polarized along xx , yy and zz directions of the electric field.

Fig. 14 presents, the real, $\varepsilon_1(\omega)$ and imaginary, $\varepsilon_2(\omega)$ parts, of the dielectric function $\varepsilon(\omega)$ calculated for pure monoclinic CuO as a function of the incident photon energy. From $\varepsilon_1(\omega)$ curves (Fig. 14 (a)), we can find that the value of the static refractive index along the xx -axis is $n_{xx}(0) = \sqrt{\text{Re}\varepsilon_{xx}(0)} = 1.98$, along the yy -axis is $n_{yy}(0) = \sqrt{\text{Re}\varepsilon_{yy}(0)} = 1.82$, and along the zz -axis is $n_{zz}(0) = \sqrt{\text{Re}\varepsilon_{zz}(0)} = 2.12$.

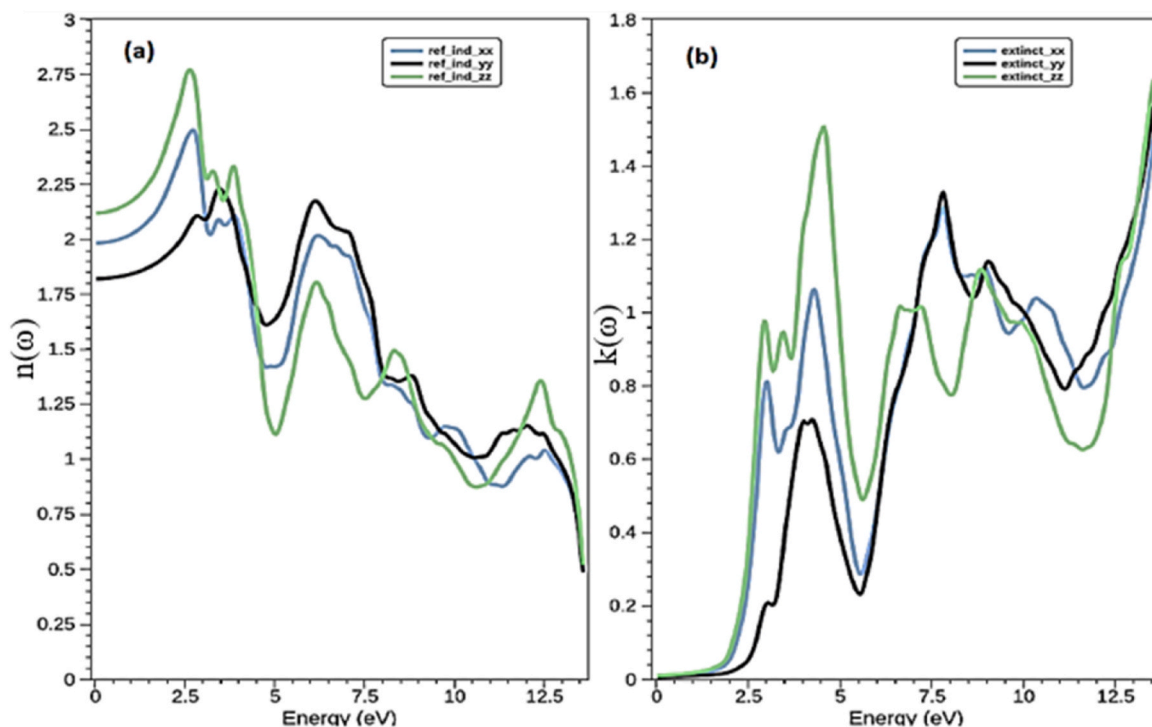


Fig. 16. The (a) refractive index and (b) Extinction coefficient of monoclinic CuO.

The $\varepsilon_2(\omega)$ curves (Fig. 14 (b)) indicate the existence of an energy threshold for all directions (xx, yy and zz) of electric field, at about 1.66 eV. This value labeled E_0 and commonly corresponds to the band gap energy; it corresponds to the absorption edge of 748 nm, indicating the response of CuO to the visible light, which probably corresponds to the intrinsic electron transition between O 2p and Cu 3d levels in the valence band maximum (VBM) and conduction band maximum (CBM), respectively. Fig. 15 (a) presents the calculated optical conductivity of CuO in terms of energy. This physical quantity can be utilized to examine the propagation and dissipation of electromagnetic waves in a material medium. From the plot, it starts between 1.5 eV and 2 eV indicating the semiconductor character of CuO.

In Fig. 15 (b) the energy loss in function of energy is presented. This physical quantity is important in characterizing the loss of energy during an electron fast traversing through a medium [49]. We observe in the curves that the energy loss for pure monoclinic CuO in all xx, yy and zz directions of electric field is weak for lower energy regime and raises with energy raising; it commences to raise between 1.5 eV and 2 eV due to the semiconductor character of CuO. The highest peaks in the range presented concern the energies of 4.72 eV and 11.50 eV for xx polarization, and of 4.61 eV and 10.54 eV for yy polarization and of 5.10 eV, 7.55 eV and 10.90 eV for zz polarization. The appearing peaks below 10 eV energy correspond to π electrons, while those appearing after corresponding to $\pi+\sigma$ plasmons. The single peaks correspond to the unique collective excitations and the numerous peaks correspond to many collective excitations [50,51].

Fig. 16 (c) displayed the reflectivity of CuO. It appears that the curves of reflectivity for all polarizations mismatch, which means that the light is polarized by this material. Besides, the curves of reflectivity of CuO exhibit sharp peaks of 20.65%, 19.83% and 23.81% at the energies of 2.84 eV, 4.23 eV and 7.80 eV respectively for xx direction, and sharp peaks of 23.81% and 20.80% at the energies of 7.80 eV and 8.94 eV respectively for yy direction, and sharp peaks of 24.89%, 29.53% and 20.79% at 2.82 eV, 4.56 eV and 8.94 eV respectively for zz direction.

Fig. 16 (a) and (b) display respectively, the refractive index ($n(\omega)$) and extinction coefficient ($k(\omega)$) spectra, in all directions of polarization. In fact, $n(\omega)$ relates to the phase velocity and refraction of an

electromagnetic wave in a medium, while $k(\omega)$ depends on the amount of absorption of a photon when it propagates in the material. Comparing Fig. 14 (a), with Fig. 16 (a), and Fig. 14 (b), with Fig. 16 (b), we remark that the variations of $n(\omega)$ and $k(\omega)$ parameters follow those of $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$ respectively. Besides, the shown values of static refractive index, $n(0)$ in Fig. 15. (a) satisfy the relation $n(0) = \sqrt{\text{Re}\varepsilon(0)}$ which indicates the rationality of our calculations.

4. Conclusion

This study examines the structural, optical, morphological, and electrical characteristics of CuO thin films that were created using spray pyrolysis and spin-coating methods. The XRD results reveal that all the films are polycrystalline, with monoclinic structure. Furthermore, the Debye-Scherrer formula yields a crystallite size between 22.07 and 29.21 nm for sprayed samples and between 23.55 and 28.73 nm for spin-coated films. Besides that, SEM images show a dense and uniform surface for sprayed films, while spin-coated images give clearly defined granule borders. The optical characteristics showed that CuO thin films developed using the spin-coating process reduce the absorbance and optical band gap of the deposited thin films. Spin coating thin films exhibit higher conductivity than sprayed thin films equal to $0.5 (\Omega \text{ m})^{-1}$. To represent the exchange and correlation functional, we used the Local-density approximation (LDA). CuO's semiconducting nature was not revealed by LDA, so we created the TB-mBJ potential. The band structure calculated using LDA-mBJ has a direct band gap energy of about 1.66 eV and is in good agreement with the experimental results. Theoretical findings from DFT theory support experimental findings and are consistent with earlier publications. The results obtained demonstrate that the elaboration method has an impact on film quality. Additionally, the results demonstrate that the prepared CuO films can be used as an absorber in photovoltaic applications.

CRediT authorship contribution statement

N. Benaissa: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing –

original draft, Writing – review & editing, Visualization, Supervision, Project administration. **H. Bentour:** Conceptualization, Methodology, Investigation, Data curation, Writing – original draft, Visualization. **T. Garmim:** Conceptualization, Methodology, Writing – review & editing, Visualization. **Z. El Jouad:** Conceptualization, Software, Investigation, Data curation, Visualization. **A. Louardi:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – review & editing, Visualization, Supervision, Project administration. **B. Hartiti:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – review & editing, Visualization, Supervision, Project administration. **M. Monkade:** Conceptualization, Software, Validation, Formal analysis, Visualization, Supervision, Project administration. **A. El Kenz:** Conceptualization, Software, Validation, Formal analysis, Visualization, Supervision, Project administration. **A. Benyoussef:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – review & editing, Visualization, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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